

		where z is the equilibrium level, with an amplitude of 1 m and period of 12.5 hours. Solving gives $t = 2.08$ or 2.1 hours.
15	C	Max $a = (2\pi \times 120)^2 \times 40 \times 10^{-3} = 22716 \text{ ms}^{-2}$ and max $v = (2\pi \times 120) \times 40 \times 10^{-3} = 30.1 \text{ ms}^{-1}$
16	D	Q leads P by $\frac{\pi}{2}$ rad. Equivalently, P leads Q by $\frac{3\pi}{2}$ rad.